

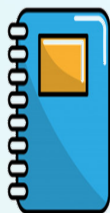
CSE 3203 – Software Engineering

Lecture 02: Introduction to SE

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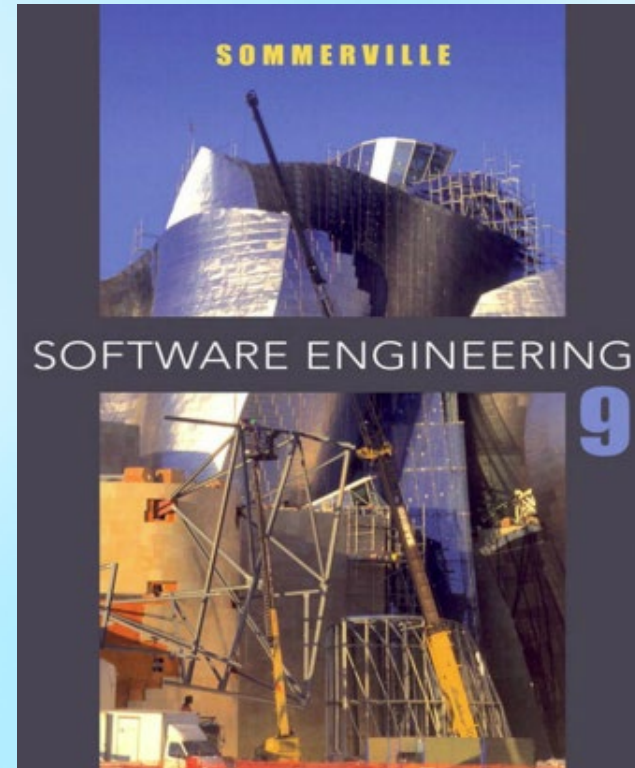
Class Code
byvb74r2



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Book for Reference

- Software Engineering 9th Edition by Ian Sommerville



Mark Distribution



- Mid Semester Exam – 25%
- Final Semester Exam – 40%
- Class Attendance & Performance – 10%
- Course Project – 25%

Is it possible to get A/A+ ?

- Absolutely. Just follow my instructions to the point.

How to fail in this course ?

- Don't follow my instructions.

Instructions



Topics covered



Professional software development

What is meant by software engineering.



Software engineering ethics

A brief introduction to ethical issues that affect software engineering.



Case studies

An introduction to three examples that are used in later chapters in the book.

Software engineering

- More and more systems are software controlled.
- Software engineering is concerned with theories, methods and tools for professional software development.
- Software costs more to maintain than it does to develop. For systems with a long life, maintenance costs may be several times development costs.

Software products

- Generic products
 - Stand-alone systems that are marketed and sold to any customer who wishes to buy them.
 - Examples – PC software such as graphics programs, project management tools; software for specific markets such as appointments systems for dentists.
- Customized products
 - Software that is commissioned by a specific customer to meet their own needs.
 - Examples – embedded control systems, air traffic control software, traffic monitoring systems.

Product specification

- Generic products
 - The specification of what the software should do is owned by the software developer and decisions on software change are made by the developer.
- Customized products
 - The specification of what the software should do is owned by the customer for the software and they make decisions on software changes that are required.

Question	Answer
What is software?	Computer programs and associated documentation. Software products may be developed for a particular customer or may be developed for a general market.
What are the attributes of good software?	Good software should deliver the required functionality and performance to the user and should be maintainable, dependable and usable.
What is software engineering?	Software engineering is an engineering discipline that is concerned with all aspects of software production.
What are the fundamental software engineering activities?	Software specification, software development, software validation and software evolution.
What is the difference between software engineering and computer science?	Computer science focuses on theory and fundamentals; software engineering is concerned with the practicalities of developing and delivering useful software.
What is the difference between software engineering and system engineering?	System engineering is concerned with all aspects of computer-based systems development including hardware, software and process engineering. Software engineering is part of this more general process.

Frequently asked questions about software engineering

Frequently asked questions about software engineering

Question	Answer
What are the key challenges facing software engineering?	Coping with increasing diversity, demands for reduced delivery times and developing trustworthy software.
What are the costs of software engineering?	Roughly 60% of software costs are development costs, 40% are testing costs. For custom software, evolution costs often exceed development costs.
What are the best software engineering techniques and methods?	While all software projects have to be professionally managed and developed, different techniques are appropriate for different types of system. For example, games should always be developed using a series of prototypes whereas safety critical control systems require a complete and analyzable specification to be developed. You can't, therefore, say that one method is better than another.
What differences has the web made to software engineering?	The web has led to the availability of software services and the possibility of developing highly distributed service-based systems. Web-based systems development has led to important advances in programming languages and software reuse.

PRODUCT CHARACTERISTIC

DESCRIPTION

Maintainability

Software should be written in such a way so that it can evolve to meet the changing needs of customers. This is a critical attribute because software change is an inevitable requirement of a changing business environment.

Dependability and security

Software dependability includes a range of characteristics including reliability, security and safety. Dependable software should not cause physical or economic damage in the event of system failure. Malicious users should not be able to access or damage the system.

Efficiency

Software should not make wasteful use of system resources such as memory and processor cycles. Efficiency therefore includes responsiveness, processing time, memory utilisation, etc.

Acceptability

Software must be acceptable to the type of users for which it is designed. This means that it must be understandable, usable and compatible with other systems that they use.

Essential attributes of good software

Software engineering

- Software engineering is an engineering discipline that is concerned with all aspects of software production from the early stages of system specification through to maintaining the system after it has gone into use.
- Engineering discipline
 - Using appropriate theories and methods to solve problems bearing in mind organizational and financial constraints.
- All aspects of software production
 - Not just technical process of development. Also project management and the development of tools, methods etc. to support software production.

Importance of software engineering

- More and more, individuals and society rely on advanced software systems. We need to be able to produce reliable and trustworthy systems economically and quickly.
- It is usually cheaper, in the long run, to use software engineering methods and techniques for software systems rather than just write the programs as if it was a personal programming project.

Software process activities

- Software specification.
- Software development.
- Software validation.
- Software evolution.

General issues that affect most software

- Heterogeneity
 - Increasingly, systems are required to operate as distributed systems across networks that include different types of computer and mobile devices.
- Business and social change
 - Business and society are changing incredibly quickly as emerging economies develop and new technologies become available.
- Security and trust
 - As software is intertwined with all aspects of our lives, it is essential that we can trust that software.



Thank You

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